



HSC  
TRIAL EXAMINATION  
2023

# Mathematics Advanced

Name: \_\_\_\_\_

Maths Class: \_\_\_\_\_

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**General  
Instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–31 show relevant mathematical reasoning and/or calculations

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**Total marks:  
100**

**Section I – 10 marks (pages 1–5)**

- Attempt Questions 1–10
- Allow about 15 minutes for this section

**Section II – 90 marks (pages 7–30)**

- Attempt Questions 11–31
- Allow about 2 hours and 45 minutes for this section

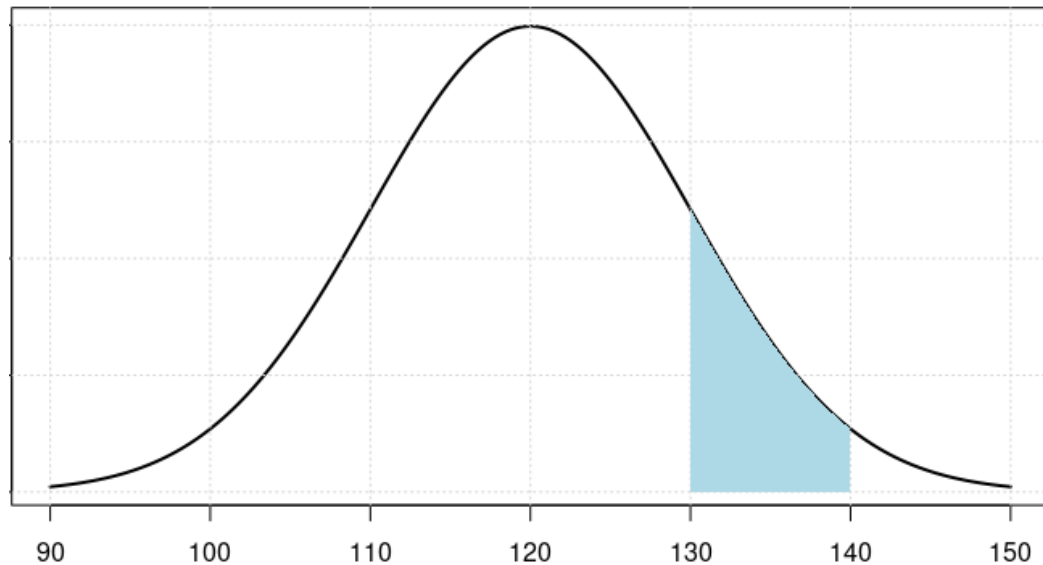
**Section I****10 marks****Attempt questions 1-10****Allow about 15 minutes for this section**Use the multiple-choice answer sheet for Questions 1–10.

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- 1 Which inequality gives the range of  $y = x^2 - 6x$ ?
- A.  $y \geq 9$
- B.  $y \geq -9$
- C.  $y \leq 9$
- D.  $y \leq -9$
- 2 Given  $x + \frac{1}{x} = 4$ , what is  $x^2 + \frac{1}{x^2}$ ?
- A. 14
- B. 16
- C. 18
- D. 20
- 3 Which of the following graphs have **no** asymptotes?
- A.  $y = \ln(x - 2)$
- B.  $y = e^{2x}$
- C.  $y = \sqrt{4 - x^2}$
- D.  $y = \frac{3}{x - 4}$

- 4 The number of cars in a car park is normally distributed with a mean of 120 and standard deviation of 10.

The shaded region represents the probability of the number of cars lying in a certain range at some time of the day.

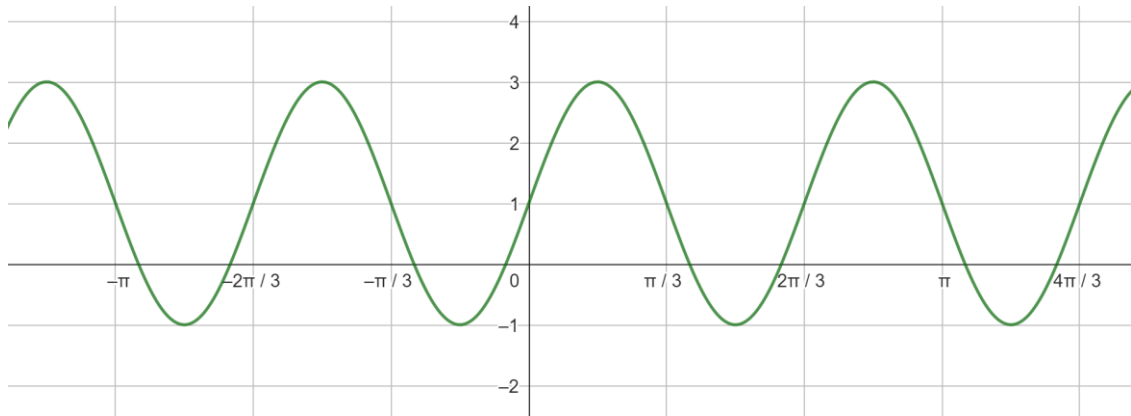


A random variable  $Z$  is normally distributed with mean 0 and standard deviation 1.

Which of the following is equivalent to the shaded area?

- A.  $P(0 < Z < 1)$
  - B.  $P(0 < Z < 2)$
  - C.  $P(-1 < Z < 0)$
  - D.  $P(-2 < Z < -1)$
- 5 What is the y-intercept of the tangent line to the curve  $y = x^3 - 5x$  at  $x = 3$ ?
- A. -54
  - B. -50
  - C. 38
  - D. 42

- 6 In the diagram, a graph of a trigonometric function is given.



Which of the following could be the equation of the given graph?

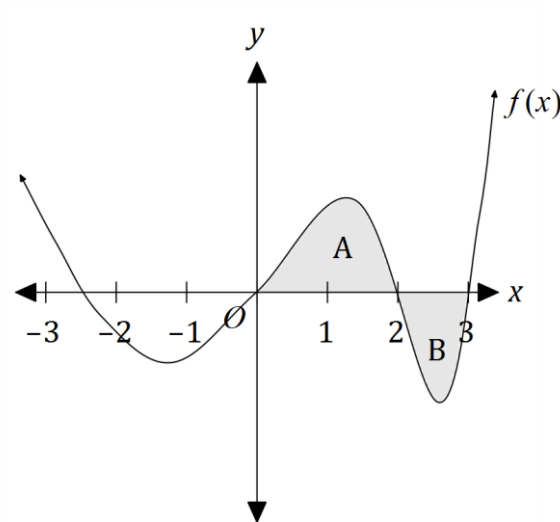
- A.  $y = 2 \sin\left(3x - \frac{\pi}{2}\right) + 1$
- B.  $y = 2 \sin(2x - \pi) + 1$
- C.  $y = 2 \cos\left(3x - \frac{\pi}{2}\right) + 1$
- D.  $y = 2 \cos\left(2x + \frac{\pi}{2}\right) + 1$
- 7 What is  $\int \frac{4}{(2x-1)^2} dx$ ?

- A.  $2 \ln(2x - 1) + c$
- B.  $\frac{-2}{2x - 1} + c$
- C.  $\frac{-2}{(2x - 1)^3} + c$
- D.  $\frac{8}{2x - 1} + c$

8 What is  $\frac{d}{dx}(x^n e^{2x})$ ?

- A.  $x^{(n-1)}(2x+n+1)e^{2x}$
- B.  $x^n(2x+n-1)e^{2x}$
- C.  $x^n(2x+n)e^{2x}$
- D.  $x^{(n-1)}(2x+n)e^{2x}$

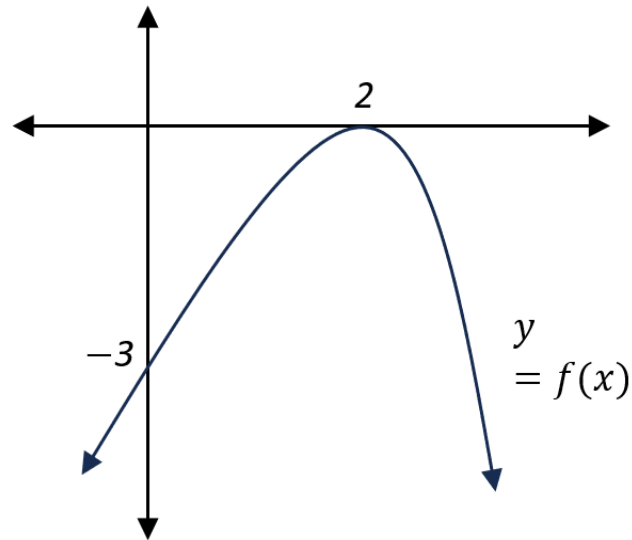
9 The graph of a function  $f(x)$  is shown below, where the shaded area labelled A is  $\frac{3}{2}$  of the shaded area labelled B.



If  $\int_0^3 f(x)dx = 2$ , what is the area of A?

- A. 6
- B. 4
- C.  $\frac{6}{5}$
- D.  $\frac{4}{5}$

- 10 A graph of the function  $f(x)$  is shown below. It has a single stationary point at  $(2,0)$  and is defined for all real  $x$ :



Define the function  $g(x) = f(f(x))$ . Which of the following is false?

- A.  $g(2) = -3$
- B.  $g(x)$  is negative for the entirety of its domain
- C.  $g'(2) = 0$
- D.  $g'(x)$  is negative for  $x < 2$

**2023**

**HSC TRIAL  
EXAMINATION**

Student Name: \_\_\_\_\_

## **Mathematics Advanced**

### **Section II**

### **Answer Booklet - I**

**30 marks**

**Attempt Questions 11–19**

**Allow about 45 minutes for this booklet**

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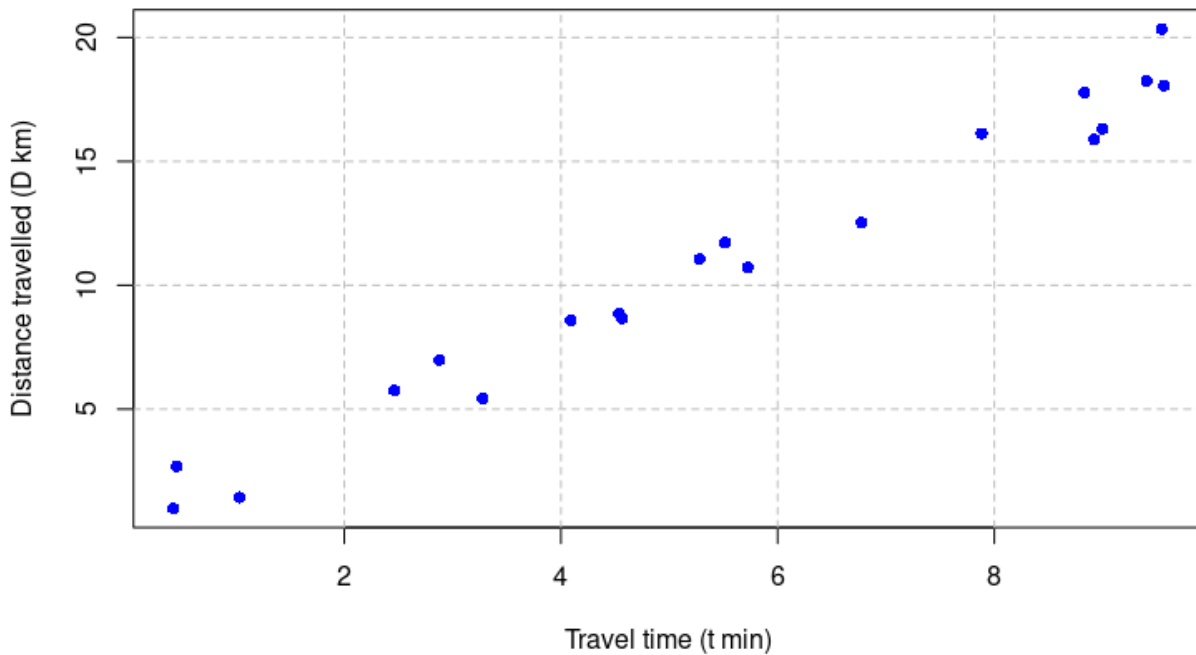
#### **Instructions**

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
  - Your responses should include relevant mathematical reasoning and/or calculations.
  - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
  - Write your name above
- 

**Please turn over**

**Question 11** (4 marks)

The relationship between the time ( $t$  minutes) taken for a taxi to travel a short distance ( $D$  kilometres) in a city centre is displayed in the following graph for 20 instances.



The equation of the least square regression line is  $D = 1.87t + 0.62$ .

- (a) State the gradient and interpret its value in the given context.

2

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- (b) On the graph above, sketch the line of best fit.

1

- (c) Use the equation of the least square regression line to find the expected travel time for a distance of 10 km? Give your answer in minutes, correct to two decimal places.

1

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**Question 12** (2 marks)

A function  $f(x)$  has derivative  $f'(x) = 4\sin(2x)$  and  $f\left(\frac{\pi}{6}\right) = 0$ .

**2**

Find  $f(x)$ .

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**Question 13** (4 marks)

(a) Differentiate  $\frac{x^2}{x^2+1}$ .

**2**


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(b) Hence, evaluate  $\int_1^2 \frac{x}{(x^2+1)^2} dx$ .

**2**


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**Question 14** (3 marks)

Describe three transformations which, when applied in succession, change the graph of  $y = x^2 + 5$  to the graph with equation  $y = \left(\frac{x+1}{2}\right)^2$ . **3**

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**Question 15** (4 marks)

Two bags contain a number of red and green marbles:

- bag A contains 3 red marbles and 4 green marbles
- bag B contains 5 red marbles and 2 green marbles.

A six-sided die is rolled. If the face of the die shows a number that is 1 or 2, then a marble from bag A will be selected at random. Otherwise, a marble from bag B will be selected.

(a) What is the probability that a green marble will be selected?

**2**


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(b) It is known that a green marble is selected.

2

What is the probability that the marble comes from bag B?

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**Question 16 (2 marks)**

Use two applications of the trapezoidal rule to approximate  $\int_1^5 \frac{12}{x^2 + 1} dx$ .

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Give your answer correct to two decimal places.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

### Question 17 (4 marks)

Sketch the graph of the curve  $y = -x^3 - 6x^2 - 9x$ , labelling the stationary points and point of inflection.

4

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**Question 18** (4 marks)

The table below gives the future value of an annuity of \$1 per period for various periods and interest rates.

| Table of Future Value Interest Factors |                          |         |         |         |         |         |         |         |
|----------------------------------------|--------------------------|---------|---------|---------|---------|---------|---------|---------|
|                                        | Interest rate per period |         |         |         |         |         |         |         |
| Number of Periods                      | 0.25%                    | 0.30%   | 0.35%   | 0.40%   | 0.45%   | 0.50%   | 0.55%   | 0.60%   |
| 53                                     | 56.5961                  | 57.3530 | 58.1230 | 58.9063 | 59.7033 | 60.5141 | 61.3391 | 62.1785 |
| 54                                     | 57.7376                  | 58.5250 | 59.3264 | 60.1419 | 60.9719 | 61.8167 | 62.6765 | 63.5516 |
| 55                                     | 58.8819                  | 59.7006 | 60.5340 | 61.3825 | 62.2463 | 63.1258 | 64.0212 | 64.9329 |
| 56                                     | 60.0291                  | 60.8797 | 61.7459 | 62.6280 | 63.5264 | 64.4414 | 65.3733 | 66.3225 |
| 57                                     | 61.1792                  | 62.0624 | 62.9620 | 63.8786 | 64.8123 | 65.7636 | 66.7329 | 67.7204 |
| 58                                     | 62.3322                  | 63.2485 | 64.1824 | 65.1341 | 66.1040 | 67.0924 | 68.0999 | 69.1267 |
| 59                                     | 63.4880                  | 64.4383 | 65.4070 | 66.3946 | 67.4014 | 68.4279 | 69.4744 | 70.5415 |
| 60                                     | 64.6467                  | 65.6316 | 66.6359 | 67.6602 | 68.7047 | 69.7700 | 70.8565 | 71.9647 |
| 61                                     | 65.8083                  | 66.8285 | 67.8692 | 68.9308 | 70.0139 | 71.1189 | 72.2463 | 73.3965 |
| 62                                     | 66.9729                  | 68.0290 | 69.1067 | 70.2065 | 71.3290 | 72.4745 | 73.6436 | 74.8369 |
| 63                                     | 68.1403                  | 69.2331 | 70.3486 | 71.4874 | 72.6499 | 73.8368 | 75.0487 | 76.2859 |
| 64                                     | 69.3106                  | 70.4408 | 71.5948 | 72.7733 | 73.9769 | 75.2060 | 76.4614 | 77.7436 |
| 65                                     | 70.4839                  | 71.6521 | 72.8454 | 74.0644 | 75.3098 | 76.5821 | 77.8820 | 79.2101 |
| 66                                     | 71.6601                  | 72.8670 | 74.1004 | 75.3607 | 76.6487 | 77.9650 | 79.3103 | 80.6854 |

- (a) Berra invests \$250 per month in an annuity which pays 5.4% p.a. compounding monthly. What will be the value of the annuity after 5 years?

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- (b) Yousif finds that he can only get an interest rate of 3.6% p.a. compounding monthly. If he wants to achieve the same total amount as Berra after the same period, what amount should he invest each month?

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**Question 19** (3 marks)Solve  $\log_2 x + \log_2(x-3) = 2$ .**3****End of Section 1I – Booklet 1**

## Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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**2023**

**HSC TRIAL  
EXAMINATION**

Student Name: \_\_\_\_\_

## **Mathematics Advanced**

### **Section II**

### **Answer Booklet - II**

**30 marks**

**Attempt Questions 20–25**

**Allow about 60 minutes for this booklet**

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#### **Instructions**

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
  - Your responses should include relevant mathematical reasoning and/or calculations.
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  - Write your name above
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**Please turn over**

### Question 20 (5 marks)

During the mushroom season, every day, Asna would go into the forest to collect mushrooms.

On the first day she collects 34 mushrooms and each consecutive day she collects 6 more mushrooms than she did on the previous day. That is, 40 mushrooms on the second day and 46 mushrooms on the third day.

- (a) On which day will Asna collect more than 120 mushrooms for the first time? 2

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- (b) During the entire mushroom season, Asna collected 3220 mushrooms. 3

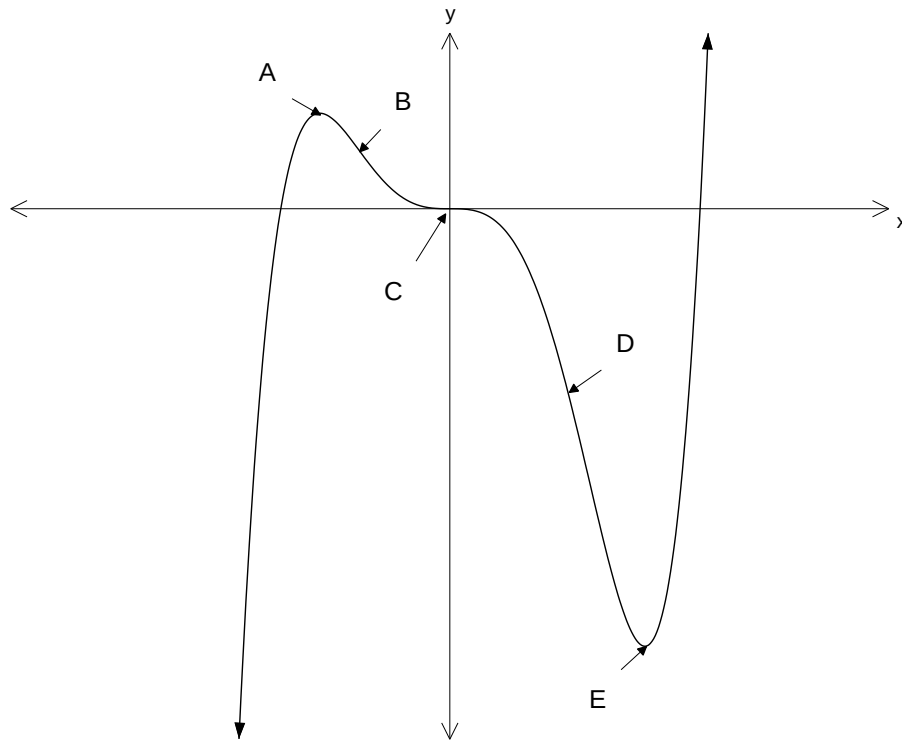
For how many days did she collect mushrooms?

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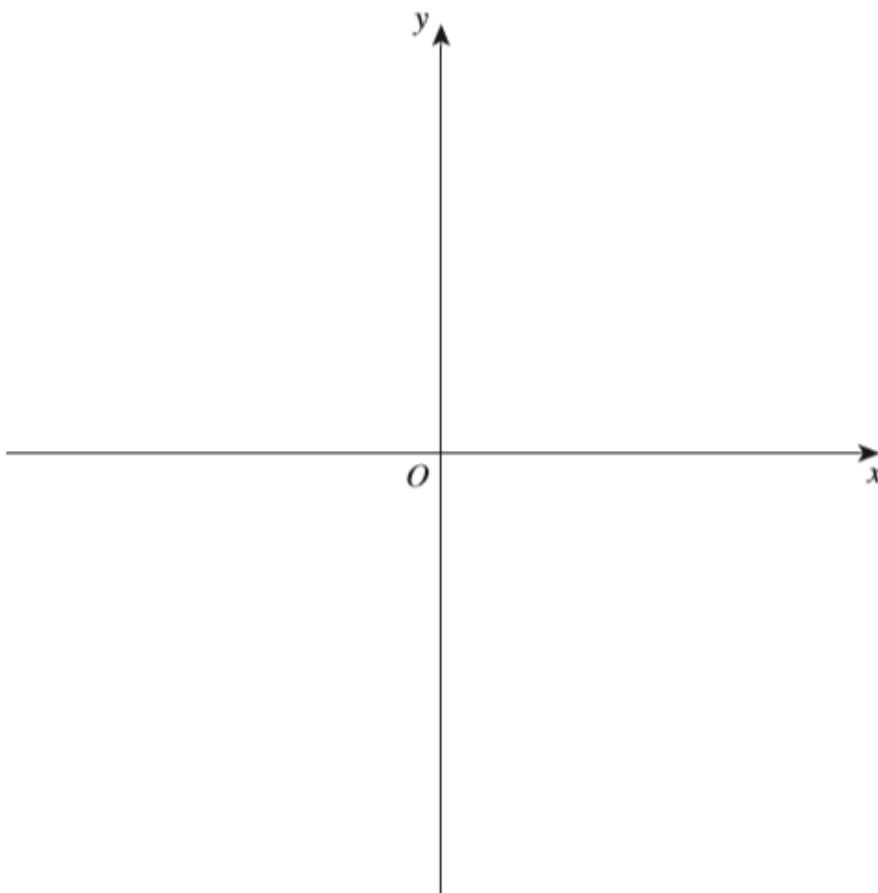
**Question 21** (2 marks)

The graph of the curve  $y = f(x)$  is drawn below:

2



On the axes below, sketch the graph of  $y = f'(x)$ , Clearly showing the behaviour of the points: A,B,C,D and E.



**Question 22** (7 marks)

The temperature,  $T$  degrees, of a liquid is given according to the model  $T(t) = 20 + Ae^{kt}$ , where  $t$  is time in hours. The initial temperature of the liquid is 35 degrees.

- (a) Show that  $A = 15$ .

**1**

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- (b) After four hours, the temperature of the liquid is 80 degrees.

**3**

Show that  $k = \ln(\sqrt{2})$ .

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**Question 22 continues on page 20**

Question 22 (continued)

- (c) Show that  $T(t) = 15 \cdot 2^{\frac{t}{2}} + 20$  and hence sketch its graph.

# 3

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**End of Question 22**

### Question 23 (9 marks)

The water at a certain beach rises and falls in a periodic pattern. The height of the water at any time can be modelled by the equation  $h(t) = 8 + 5 \sin\left(\frac{\pi t}{12}\right)$ , where  $t$  is time in hours after midnight.

- (a) Sketch the graph of  $y = h(t)$  for  $0 \leq t \leq 24$ .

2

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- (b) At what rate does the height of water change at 8:00 am? Leave your answer in exact form.

2

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**Question 23 continues on page 22**

Question 23 (continued)

- (c) A boat with a depth of 5.5 metres, is scheduled to arrive at the beach at 10:00 am one day. 2

Explain if the boat will be able to dock.

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- (d) On each day, between what times will the boat in part (c) be able to dock? 3

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**End of Question 23**

**Question 24 (3 marks)**

Find the value of  $k$  such that  $\int_0^k e^x(e^x - 2)dx = \frac{3}{2}$ .

# 3

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### Question 25 (4 marks)

A continuous random variable  $X$  has probability density function given by:

$$f(x) = \begin{cases} \frac{x+2}{k} & 1 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

- (a) Show that  $k = 8$ .

# 1

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

- (b) Find  $P(X < 2 | X > 1.5)$ .

3

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**End of Section 1I – Booklet II**

## Section II extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

**2023**

**HSC TRIAL  
EXAMINATION**

Student Name: \_\_\_\_\_

## **Mathematics Advanced**

### **Section II**

### **Answer Booklet - III**

**30 marks**

**Attempt Questions 26–31**

**Allow about 60 minutes for this booklet**

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#### **Instructions**

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
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  - Write your name above
- 

**Please turn over**

### Question 26 (5 marks)

A random variable is normally distributed with mean 0 and standard deviation 1.

The table gives the probability that this random variable is less than  $z$  for different values of  $z$ .

|            |        |        |        |        |        |        |        |        |        |
|------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| $z$        | -2     | -1.5   | -1     | -0.5   | 0      | 0.5    | 1      | 1.5    | 2      |
| $P(Z < z)$ | 0.0228 | 0.0668 | 0.1587 | 0.3085 | 0.5000 | 0.6915 | 0.8413 | 0.9332 | 0.9772 |

- (a) Using the table, find the probability that a value from a random variable that is normally distributed with mean 0 and standard deviation 1, lies between  $-0.5$  and  $1$ .

# 1

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- (b) The blood pressure readings of patients with a certain condition are normally distributed, with a mean of  $\mu$  and a standard deviation of  $\sigma$ . It is known that 2.28% of these patients have a blood reading of less than 105 and 6.68% of these patients have a blood reading of more than 133.

4

Using the table, find  $\mu$  and  $\sigma$ .

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### Question 27 (4 marks)

Find the global maximum and minimum values of  $f(x) = \sin^2 x + 4 \cos x$ , where  $\frac{\pi}{3} \leq x \leq \frac{3\pi}{2}$ .

4

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

### Question 28 (5 marks)

A container in the shape of a right circular cylinder with no top, has outside surface area  $12\pi$  square metres.

- (a) Show that the height ( $h$ ) is  $\frac{12-r^2}{2r}$ , where  $r$  is the radius.

# 1

[illegible]

- (b) Find and verify the height ( $h$ ) and radius ( $r$ ) that give the maximum volume.

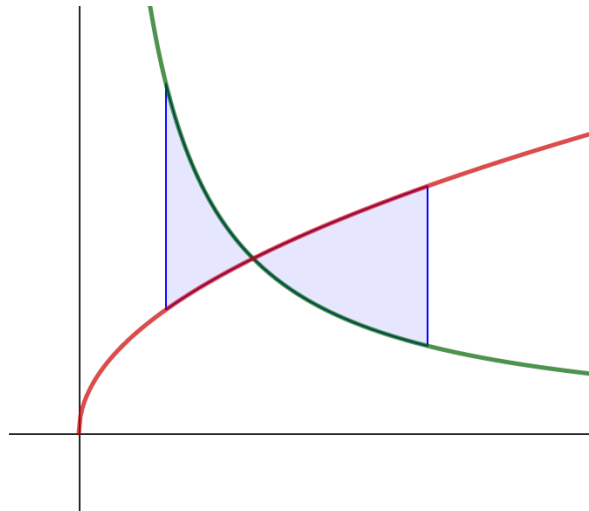
4

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

### Question 29 (4 marks)

In the diagram, the region between the graphs of  $y = \frac{4}{x}$  and  $y = \sqrt{2x}$  for  $1 \leq x \leq 4$  is shaded.

4



Find the exact area of the shaded region.

This image shows a full page of a document template designed for writing. It features a series of evenly spaced, light gray horizontal lines that run across the entire width of the page. The background is a solid off-white color. There are no margins, headers, footers, or other markings present on the page.

### Question 30 (8 marks)

Eman invests \$9000 in an account earning interest at a rate of 0.6% per month. Each month, immediately after the interest has been paid, Eman contributes \$200.

The amount in the account after the  $n$ th contribution can be determined using the recurrence relation  $A_n = 1.006A_{n-1} + 200$ , where  $A_0 = 9000$ .

- (a) Find the amount in the account after the first contribution.

# 1

[illegible]

- (b) Find the total value of the investment after the 12<sup>th</sup> contribution.

3

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

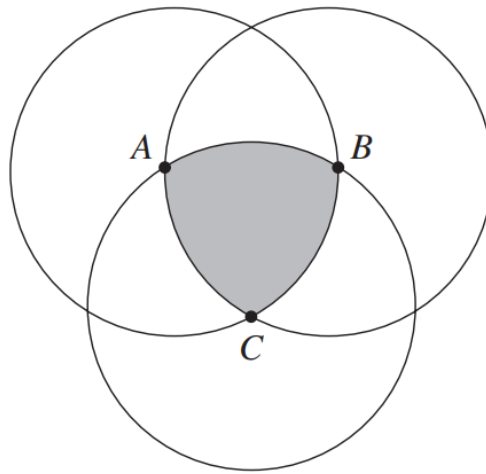
Question 30 continues on page 32



### Question 31 (4 marks)

Three circles with radius  $r$  are drawn such that their centres lie on the circumference of the other two circles, as shown in the diagram.

4



Points  $A$ ,  $B$  and  $C$  are the centres of the three circles. Prove that the area of the shaded region is

$$\frac{1}{2}r^2(\pi - \sqrt{3}).$$

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**End of EXAM**

## Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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## Section II extra writing space

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**2023****HIGHER SCHOOL CERTIFICATE  
TRIAL EXAMINATION**

Student Name: \_\_\_\_\_

**Mathematics Advanced Yr 12**  
**Section I Answer Sheet****10 marks****Attempt Questions 1–10****Allow about 15 minutes for this section**

Select the alternative A, B, C or D that best answers the question. Fill in the response circle completely.

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- |           |                         |                         |                         |                         |
|-----------|-------------------------|-------------------------|-------------------------|-------------------------|
| <b>1</b>  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| <b>2</b>  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| <b>3</b>  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| <b>4</b>  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| <b>5</b>  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| <b>6</b>  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| <b>7</b>  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| <b>8</b>  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| <b>9</b>  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| <b>10</b> | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |

Mathematics Advanced  
Mathematics Extension 1  
Mathematics Extension 2

## REFERENCE SHEET

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### Measurement

#### Length

$$l = \frac{\theta}{360} \times 2\pi r$$

#### Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a+b)$$

#### Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

#### Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

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### Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\text{For } ax^3 + bx^2 + cx + d = 0:$$

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

#### Relations

$$(x-h)^2 + (y-k)^2 = r^2$$

### Financial Mathematics

$$A = P(1+r)^n$$

#### Sequences and series

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a+l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{a(1 - r^n)}{1 - r}, r \neq 1$$

$$S = \frac{a}{1-r}, |r| < 1$$

---

### Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

## Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2} ab \sin C$$

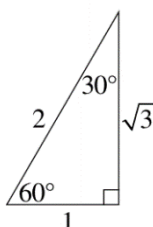
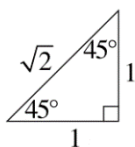
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2} r^2 \theta$$



## Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

## Compound angles

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A+B) - \sin(A-B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

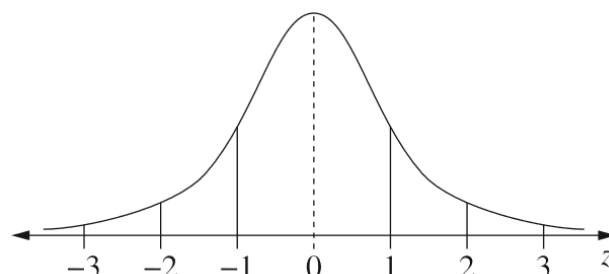
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

## Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score  
less than  $Q_1 - 1.5 \times IQR$   
or  
more than  $Q_3 + 1.5 \times IQR$

## Normal distribution



- approximately 68% of scores have z-scores between  $-1$  and  $1$
- approximately 95% of scores have z-scores between  $-2$  and  $2$
- approximately 99.7% of scores have z-scores between  $-3$  and  $3$

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

## Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

## Continuous random variables

$$P(X \leq x) = \int_a^x f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

## Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

## Differential Calculus

### Function

### Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = nf'(x)[f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u), u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1-[f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1-[f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1+[f(x)]^2}$$

## Integral Calculus

$$\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + c$$

where  $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx \approx \frac{b-a}{2n} \{ f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})] \}$$

where  $a = x_0$  and  $b = x_n$

## Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \dots + \binom{n}{r}x^{n-r}a^r + \dots + a^n$$


---

## Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}||\underline{v}|\cos\theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = a + \lambda b$$


---

## Complex Numbers

$$z = a + ib = r(\cos\theta + i\sin\theta) \\ = re^{i\theta}$$

$$\left[r(\cos\theta + i\sin\theta)\right]^n = r^n(\cos n\theta + i\sin n\theta) \\ = r^n e^{in\theta}$$


---

## Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$





HSC  
TRIAL EXAMINATION  
2023

# Mathematics Advanced

Name: Solutions

Maths Class: \_\_\_\_\_

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General  
Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–31 show relevant mathematical reasoning and/or calculations

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Total marks:  
100

Section I – 10 marks (pages 1–5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7–30)

- Attempt Questions 11–31
- Allow about 2 hours and 45 minutes for this section

## Section I

10 marks

Attempt questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Which inequality gives the range of  $y = x^2 - 6x$ ?

A.  $y \geq 9$

B.  $y \geq -9$

C.  $y \leq 9$

☒ D.  $y \leq -9$

$y = x^2 - 6x$ . Concave up

tp (3, -9) 

$\therefore y \geq -9$

2 Given  $x + \frac{1}{x} = 4$ , what is  $x^2 + \frac{1}{x^2}$ ?

☒ A. 14

B. 16

C. 18

D. 20

$(x + \frac{1}{x})^2 = 4^2$

$x^2 + 2 \cdot x \cdot \frac{1}{x} + \frac{1}{x^2} = 16$

$x^2 + \frac{1}{x^2} = 14$

3 Which of the following graphs have no asymptotes?

A.  $y = \ln(x - 2)$   $\rightarrow x = 2$

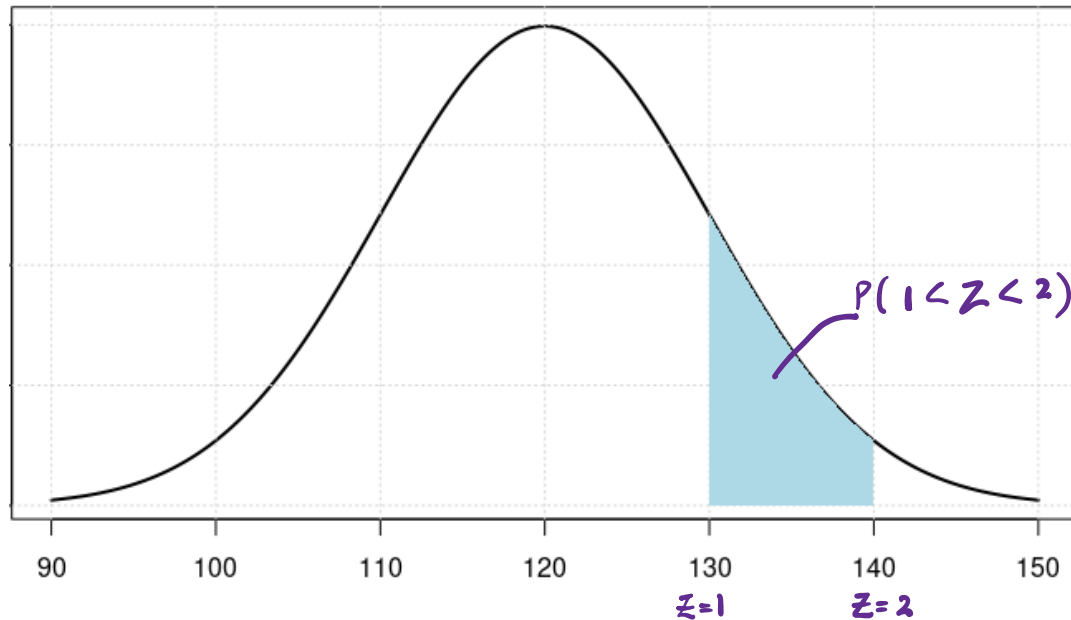
B.  $y = e^{2x}$   $\rightarrow y = 0$

☒ C.  $y = \sqrt{4 - x^2}$   $\rightarrow$  semicircle no asymptote.

D.  $y = \frac{3}{x - 4}$   $\rightarrow x = 4, y = 0$

- 4 The number of cars in a car park is normally distributed with a mean of 120 and standard deviation of 10.

The shaded region represents the probability of the number of cars lying in a certain range at some time of the day.



A random variable  $Z$  is normally distributed with mean 0 and standard deviation 1.

Which of the following is equivalent to the shaded area?

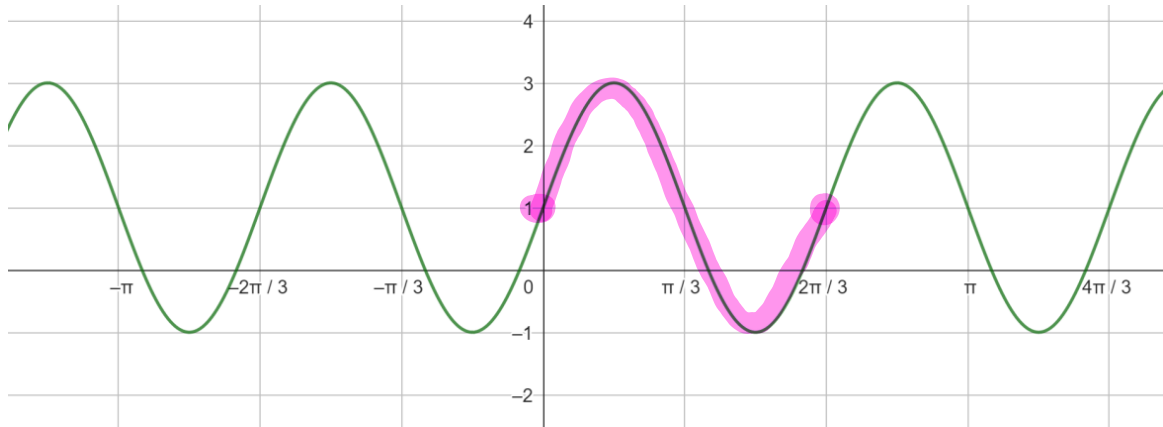
- A.  $P(0 < Z < 1)$
- B.  $P(0 < Z < 2)$
- C.  $P(-1 < Z < 0)$
- ☒ D.  $P(-2 < Z < -1)$

- 5 What is the y-intercept of the tangent line to the curve  $y = x^3 - 5x$  at  $x = 3$ ?

- ☒ A. -54
- B. -50
- C. 38
- D. 42

$$\begin{aligned}
 y &= x^3 - 5x \\
 y(3) &= 12 \\
 y' &= 3x^2 - 5 \\
 \text{at } x &= 3 \\
 y'(3) &= 3 \cdot 3^2 - 5 = 22 \\
 m_{\text{tangent}} &= 22 \\
 y - 12 &= 22(x - 3) \\
 y &= 22x - 66 + 12 \\
 y &= 22x - 54
 \end{aligned}$$

- 6 In the diagram, a graph of a trigonometric function is given.



Which of the following could be the equation of the given graph?

A.  $y = 2 \sin\left(3x - \frac{\pi}{2}\right) + 1$

B.  $y = 2 \sin(2x - \pi) + 1$

**C.**  $y = 2 \cos\left(3x - \frac{\pi}{2}\right) + 1$

D.  $y = 2 \cos\left(2x + \frac{\pi}{2}\right) + 1$

period =  $\frac{2\pi}{3}$

range =  $[-1, 3]$

check y-int when  $x=0$

$$y = 2 \cos\left(3 \cdot 0 - \frac{\pi}{2}\right) + 1$$

$$y = 2(0) + 1$$

$$y = 1$$

- 7 What is  $\int \frac{4}{(2x-1)^2} dx$ ?

A.  $2 \ln(2x - 1) + c$

**B.**  $\frac{-2}{2x-1} + c$

C.  $\frac{-2}{(2x-1)^3} + c$

D.  $\frac{8}{2x-1} + c$

$$\int 4(2x-1)^{-2} dx$$

$$= \frac{4(2x-1)^{-1}}{-1 \times 2} + c$$

$$= \frac{-2}{(2x-1)} + c$$

8 What is  $\frac{d}{dx}(x^n e^{2x})$ ?

A.  $x^{(n-1)}(2x+n+1)e^{2x}$

B.  $x^n(2x+n-1)e^{2x}$

C.  $x^n(2x+n)e^{2x}$

**D.**  $x^{(n-1)}(2x+n)e^{2x}$

$$u = x^n \quad v = e^{2x}$$

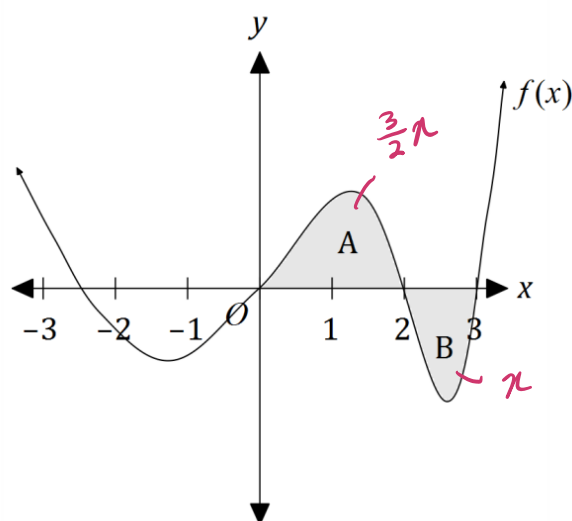
$$u' = nx^{n-1} \quad v' = 2e^{2x}$$

$$y' = 2 \cdot x^n e^{2x} + n \cdot x^{n-1} e^{2x}$$

$$y' = x^{n-1} e^{2x} (2 \cdot x + n)$$

$$y' = x^{n-1} (2x+n) \cdot e^{2x}$$

9 The graph of a function  $f(x)$  is shown below, where the shaded area labelled A is  $\frac{3}{2}$  of the shaded area labelled B.



If  $\int_0^3 f(x) dx = 2$ , what is the area of A?

**A.** 6

B. 4

C.  $\frac{6}{5}$

D.  $\frac{4}{5}$

$$\int_0^3 f(x) dx = 2$$

$$B = \pi$$

$$A = \frac{3}{2} \pi$$

$$\frac{3}{2} \pi - \pi = 2$$

$$\frac{1}{2} \pi = 2$$

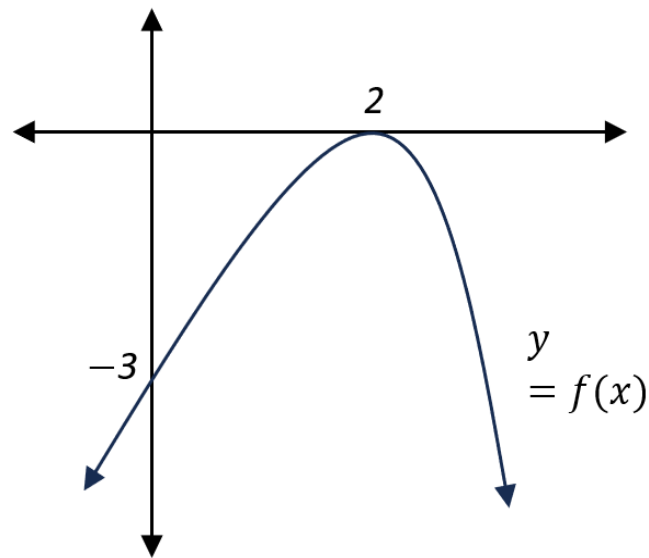
$$\pi = 4$$

$$A = \frac{3}{2} \pi$$

$$= \frac{3}{2} \times 4$$

$$= 6$$

- 10 A graph of the function  $f(x)$  is shown below. It has a single stationary point at  $(2,0)$  and is defined for all real  $x$ :



Define the function  $g(x) = f(f(x))$ . Which of the following is false?

- A.  $g(2) = -3$
- B.  $g(x)$  is negative for the entirety of its domain
- C.  $g'(2) = 0$
- D.  $g'(x)$  is negative for  $x < 2$**

WS

Student Name: \_\_\_\_\_

**2023**

**HSC TRIAL  
EXAMINATION**

## Mathematics Advanced

### Section II

### Answer Booklet - I

**30 marks**

**Attempt Questions 11–19**

**Allow about 45 minutes for this booklet**

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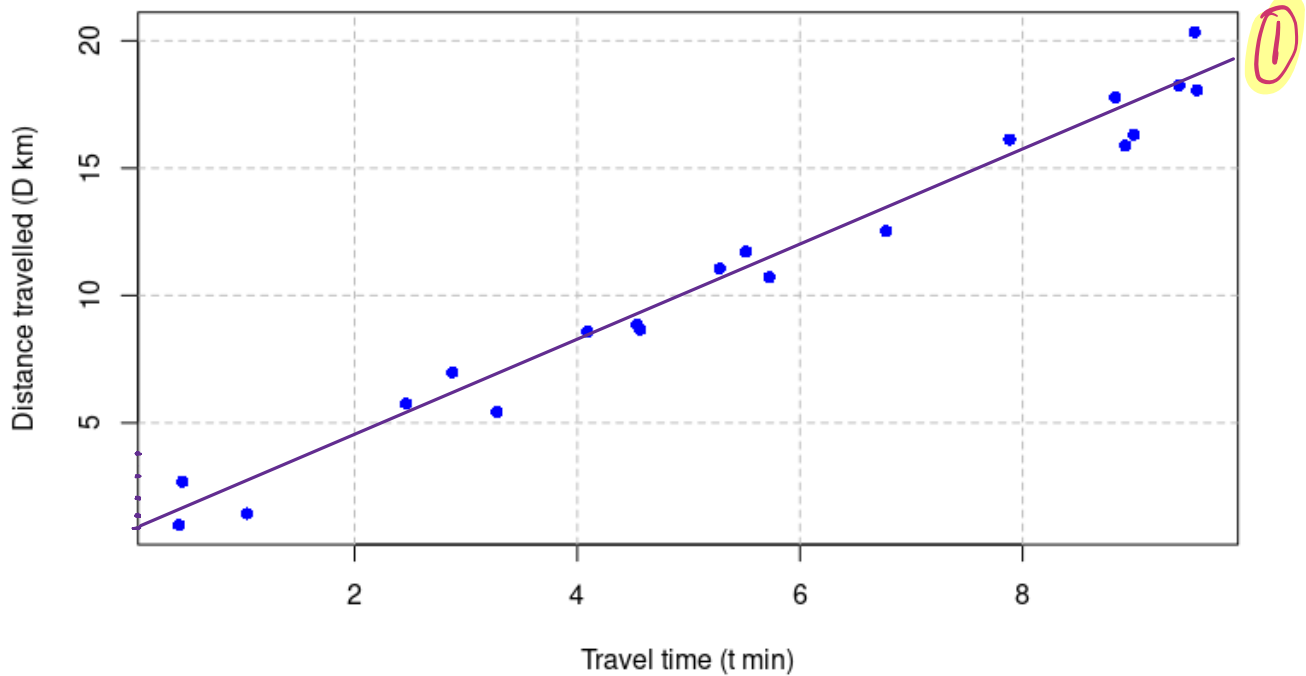
#### Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
  - Your responses should include relevant mathematical reasoning and/or calculations.
  - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
  - Write your name above
- 

**Please turn over**

**Question 11** (4 marks)

The relationship between the time ( $t$  minutes) taken for a taxi to travel a short distance ( $D$  kilometres) in a city centre is displayed in the following graph for 20 instances.



The equation of the least square regression line is  $D = 1.87t + 0.62$ .

- (a) State the gradient and interpret its value in the given context.

2

$m = 1.87$  ①  
 • for every 1 min increase in time distance  
 or Increase by 1.87 km ①  
 • Distance travelled is almost double the travel time.

- (b) On the graph above, sketch the line of best fit.

1

- (c) Use the equation of the least square regression line to find the expected travel time for a distance of 10 km? Give your answer in minutes, correct to two decimal places.

1

$D = 1.87t + 0.62$   
 $10 = 1.87t + 0.62$   
 $t \doteq 5.016042781$   
 $t \approx 5.02 \text{ mins.}$  ①

**Question 12** (2 marks)

A function  $f(x)$  has derivative  $f'(x) = 4\sin(2x)$  and  $f\left(\frac{\pi}{6}\right) = 0$ .

2

Find  $f(x)$ .

$$\begin{aligned}
 f(x) &= \int 4\sin(2x) \, dx \\
 &= -\frac{4^2 \cos(2x)}{2} + C \\
 &= -2\cos(2x) + C \\
 f\left(\frac{\pi}{6}\right) &= 0 \\
 0 &= -2\cos\left(2 \cdot \frac{\pi}{6}\right) + C \\
 0 &= -2\cos\left(\frac{\pi}{3}\right) + C \\
 0 &= -2\left(\frac{1}{2}\right) + C \\
 C &= 1 \\
 \therefore f(x) &= -2\cos(2x) + 1
 \end{aligned}$$

**Question 13** (4 marks)

(a) Differentiate  $\frac{x^2}{x^2+1}$ .

2

$$\begin{aligned}
 u &= x^2 & v &= x^2 + 1 \\
 u' &= 2x & v' &= 2x \\
 \frac{2x(x^2+1) - 2x \cdot x^2}{(x^2+1)^2} &= \frac{2x^3 + 2x - 2x^3}{(x^2+1)^2} \\
 &= \frac{2x}{(x^2+1)^2}
 \end{aligned}$$

(b) Hence, evaluate  $\int_1^2 \frac{x}{(x^2+1)^2} \, dx$ .

2

$$\begin{aligned}
 &= \frac{1}{2} \int_1^2 \frac{2x}{(x^2+1)^2} \, dx \\
 &= \frac{1}{2} \left[ \frac{x^2}{x^2+1} \right]_1^2 \\
 &= \frac{1}{2} \left[ \frac{2^2}{2^2+1} - \frac{1^2}{1^2+1} \right] \\
 &= \frac{1}{2} \left( \frac{4}{5} - \frac{1}{2} \right) \\
 &= \frac{3}{20}
 \end{aligned}$$

**Question 14** (3 marks)

Describe three transformations which, when applied in succession, change the graph of  $y = x^2 + 5$  to the graph with equation  $y = \left(\frac{x+1}{2}\right)^2$ . 3

$$y = x^2 + 5 \rightarrow x^2 \rightarrow \left(\frac{x}{2}\right)^2 \rightarrow \left(\frac{x+1}{2}\right)^2$$

- 5 units down
- H. Dilation by factor of 2
- 1 unit left

OR:  $y = \left(\frac{x}{2} + \frac{1}{2}\right)^2$

- $5 \downarrow, \leftarrow \frac{1}{2}, 2 \times \text{H.D.}$

OR:  $\downarrow 5, \leftarrow 1, D \times \frac{1}{4} \text{ vertically}$

OR:  $\downarrow 5, D \times \frac{1}{4}, \leftarrow 1$

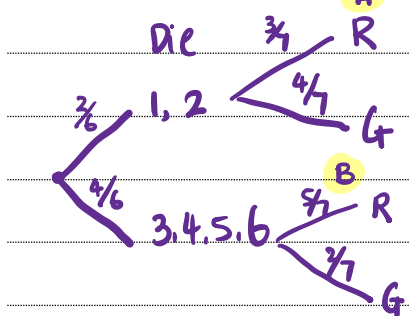
**Question 15** (4 marks)

Two bags contain a number of red and green marbles:

- bag A contains 3 red marbles and 4 green marbles
- bag B contains 5 red marbles and 2 green marbles.

A six-sided die is rolled. If the face of the die shows a number that is 1 or 2, then a marble from bag A will be selected at random. Otherwise, a marble from bag B will be selected.

(a) What is the probability that a green marble will be selected? 2



$$P(\text{Green}) = P(D_{1,2}, A_G) + P(D_{3,4,5,6}, B_G)$$

$$= \left(\frac{2}{6} \times \frac{4}{7}\right) + \left(\frac{4}{6} \times \frac{2}{7}\right)$$

$$= \frac{8}{21}$$

- (b) It is known that a green marble is selected.

2

What is the probability that the marble comes from bag B?

$$\begin{aligned}
 P(B/G) &= \frac{P(B \cap G)}{P(G)} \\
 &= \frac{\frac{4}{6} \times \frac{2}{7}}{\frac{8}{21}} \quad \textcircled{1} \\
 &= \frac{\frac{8}{42}}{\frac{8}{21}} \\
 &= \frac{1}{2} \quad \textcircled{1}
 \end{aligned}$$

**Question 16** (2 marks)Use two applications of the trapezoidal rule to approximate  $\int_1^5 \frac{12}{x^2+1} dx$ .

2

Give your answer correct to two decimal places.

| $x$    | 1                  | 3                  | 5                  |
|--------|--------------------|--------------------|--------------------|
| $f(x)$ | 6                  | $\frac{12}{10}$    | $\frac{12}{26}$    |
|        | $\frac{12}{1^2+1}$ | $\frac{12}{3^2+1}$ | $\frac{12}{5^2+1}$ |
|        | = 6                | = 1.2              |                    |

$$\begin{aligned}
 &\int_1^5 \frac{12}{x^2+1} dx \\
 &= \frac{h}{2} [y_0 + 2y_1 + y_2] \\
 &= \frac{2}{2} \left[ 6 + 2 \times \frac{12}{10} + \frac{12}{26} \right] \quad \textcircled{1} \\
 &\div 8.86 \quad \textcircled{1}
 \end{aligned}$$

## Question 17 (4 marks)

Sketch the graph of the curve  $y = -x^3 - 6x^2 - 9x$ , labelling the stationary points and point of inflection.

4

$$y = -x^3 - 6x^2 - 9x$$

$$y' = -3x^2 - 12x - 9$$

$$y' = 0 \rightarrow \text{stat pt}$$

$$0 = -3x^2 - 12x - 9$$

$$0 = x^2 + 4x + 3$$

$$0 = (x+3)(x+1)$$

$$\textcircled{1} \begin{pmatrix} x = -1 \\ y = 4 \end{pmatrix} \text{ or } \begin{pmatrix} x = -3 \\ y = 0 \end{pmatrix} \textcircled{1}$$

$\wedge$  max  
tp

$\vee$  min  
tp

$$y'' = -6x - 12$$

$$y'' = 0 \rightarrow \text{inf pts}$$

$$0 = -6x - 12$$

$$0 = x + 2$$

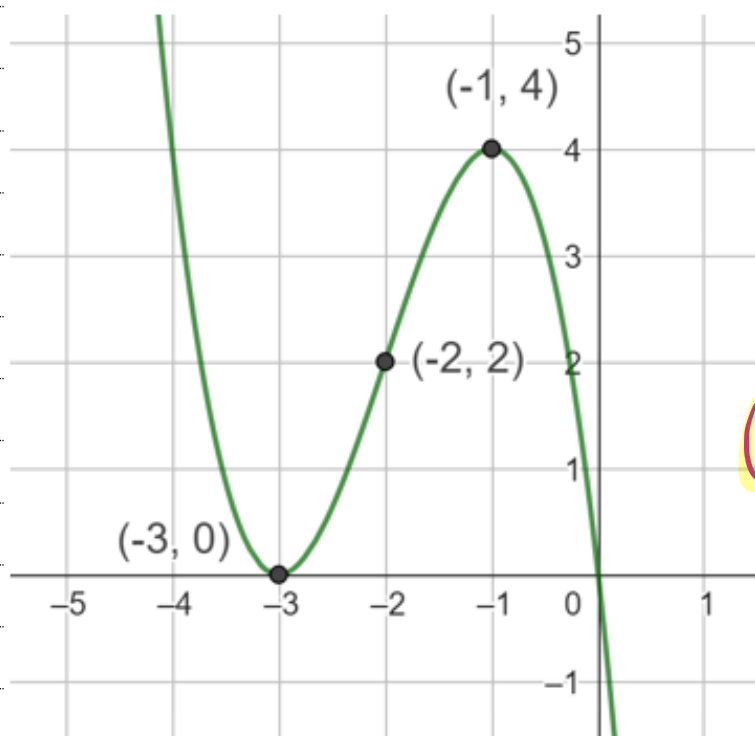
$$\begin{pmatrix} x = -2 \\ y = 2 \end{pmatrix} \text{ inf pt } \textcircled{1}$$

$$y''(-1) < 0 \wedge \text{max}$$

$$y''(-3) > 0 \vee \text{min}$$

test  $x = -2$

| $x$   | -2.1 | -2 | -1.9 |
|-------|------|----|------|
| $y''$ | +    | 0  | -    |



$\textcircled{1}$

Question 18 (4 marks)

The table below gives the future value of an annuity of \$1 per period for various periods and interest rates.

| Table of Future Value Interest Factors |                          |         |         |         |         |         |         |         |
|----------------------------------------|--------------------------|---------|---------|---------|---------|---------|---------|---------|
|                                        | Interest rate per period |         |         |         |         |         |         |         |
| Number of Periods                      | 0.25%                    | 0.30%   | 0.35%   | 0.40%   | 0.45%   | 0.50%   | 0.55%   | 0.60%   |
| 53                                     | 56.5961                  | 57.3530 | 58.1230 | 58.9063 | 59.7033 | 60.5141 | 61.3391 | 62.1785 |
| 54                                     | 57.7376                  | 58.5250 | 59.3264 | 60.1419 | 60.9719 | 61.8167 | 62.6765 | 63.5516 |
| 55                                     | 58.8819                  | 59.7006 | 60.5340 | 61.3825 | 62.2463 | 63.1258 | 64.0212 | 64.9329 |
| 56                                     | 60.0291                  | 60.8797 | 61.7459 | 62.6280 | 63.5264 | 64.4414 | 65.3733 | 66.3225 |
| 57                                     | 61.1792                  | 62.0624 | 62.9620 | 63.8786 | 64.8123 | 65.7636 | 66.7329 | 67.7204 |
| 58                                     | 62.3322                  | 63.2485 | 64.1824 | 65.1341 | 66.1040 | 67.0924 | 68.0999 | 69.1267 |
| 59                                     | 63.4880                  | 64.4383 | 65.4070 | 66.3946 | 67.4014 | 68.4279 | 69.4744 | 70.5415 |
| 60                                     | 64.6467                  | 65.6316 | 66.6359 | 67.6602 | 68.7047 | 69.7700 | 70.8565 | 71.9647 |
| 61                                     | 65.8083                  | 66.8285 | 67.8692 | 68.9308 | 70.0139 | 71.1189 | 72.2463 | 73.3965 |
| 62                                     | 66.9729                  | 68.0290 | 69.1067 | 70.2065 | 71.3290 | 72.4745 | 73.6436 | 74.8369 |
| 63                                     | 68.1403                  | 69.2331 | 70.3486 | 71.4874 | 72.6499 | 73.8368 | 75.0487 | 76.2859 |
| 64                                     | 69.3106                  | 70.4408 | 71.5948 | 72.7733 | 73.9769 | 75.2060 | 76.4614 | 77.7436 |
| 65                                     | 70.4839                  | 71.6521 | 72.8454 | 74.0644 | 75.3098 | 76.5821 | 77.8820 | 79.2101 |
| 66                                     | 71.6601                  | 72.8670 | 74.1004 | 75.3607 | 76.6487 | 77.9650 | 79.3103 | 80.6854 |

- (a)

Berra invests \$250 per month in an annuity which pays 5.4% p.a. compounding monthly. What will be the value of the annuity after 5 years?

2

$$\frac{5.4\%}{12} = 0.45\% \text{ per month}$$
$$5\text{ yr} = 5 \times 12 = 60 \text{ month}$$
$$\text{Factor} = 68.7047$$

$$FV = 68.7047 \times 250$$
$$= \$17176.18$$

- (b)

Yousif finds that he can only get an interest rate of 3.6% p.a. compounding monthly. If he wants to achieve the same total amount as Berra after the same period, what amount should he invest each month?

2

$$\frac{3.6\%}{12} = 0.30\% \text{ per month}$$
$$5\text{ yr} = 5 \times 12 = 60 \text{ mon}$$
$$\text{Factor} = 65.6316$$

$$17176.18 = x \times 65.6316$$
$$\$261.71 = x$$

## Question 19 (3 marks)

Solve  $\log_2 x + \log_2(x-3) = 2$ .

3

$$x > 0 \text{ \& \> } x-3 > 0$$

$$\boxed{x > 3}$$

$$\log_2 x + \log_2(x-3) = 2$$

$$\log_2 x \cdot (x-3) = 2$$

$$2^2 = x^2 - 3x$$

(1)

$$0 = x^2 - 3x - 4$$

$$0 = (x-4)(x+1)$$

$$x = 4 \text{ or } x = -1$$

(1)

not

valid as  $x > 3$ 

(1)

$$\therefore x = 4$$

End of Section 1I – Booklet 1

## Section II extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

**2023**

**HSC TRIAL  
EXAMINATION**

Student Name: \_\_\_\_\_

## **Mathematics Advanced**

### **Section II**

### **Answer Booklet - II**

**30 marks**

**Attempt Questions 20–25**

**Allow about 60 minutes for this booklet**

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**Instructions**

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
  - Your responses should include relevant mathematical reasoning and/or calculations.
  - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
  - Write your name above
- 

**Please turn over**

**Question 20** (5 marks)

During the mushroom season, every day, Asna would go into the forest to collect mushrooms.

On the first day she collects 34 mushrooms and each consecutive day she collects 6 more mushrooms than she did on the previous day. That is, 40 mushrooms on the second day and 46 mushrooms on the third day.

- (a) On which day will Asna collect more than 120 mushrooms for the first time?

2

$$\begin{aligned}
 &34, 40, 46 \dots \\
 &a_1 = 34 \qquad 34 + (n-1) \cdot 6 > 120 \quad (1) \\
 &d = 6 \qquad 34 + 6n - 6 > 120 \\
 &a_n > 120 \qquad 6n > 92 \\
 &\qquad \qquad n > \frac{92}{6} \\
 &\qquad \qquad n > 15.3 \dots \quad \rightarrow n = 16^{\text{th}} \text{ day} \quad (1) \\
 &a_n = a_1 + (n-1)d
 \end{aligned}$$

- (b) During the entire mushroom season, Asna collected 3220 mushrooms.

3

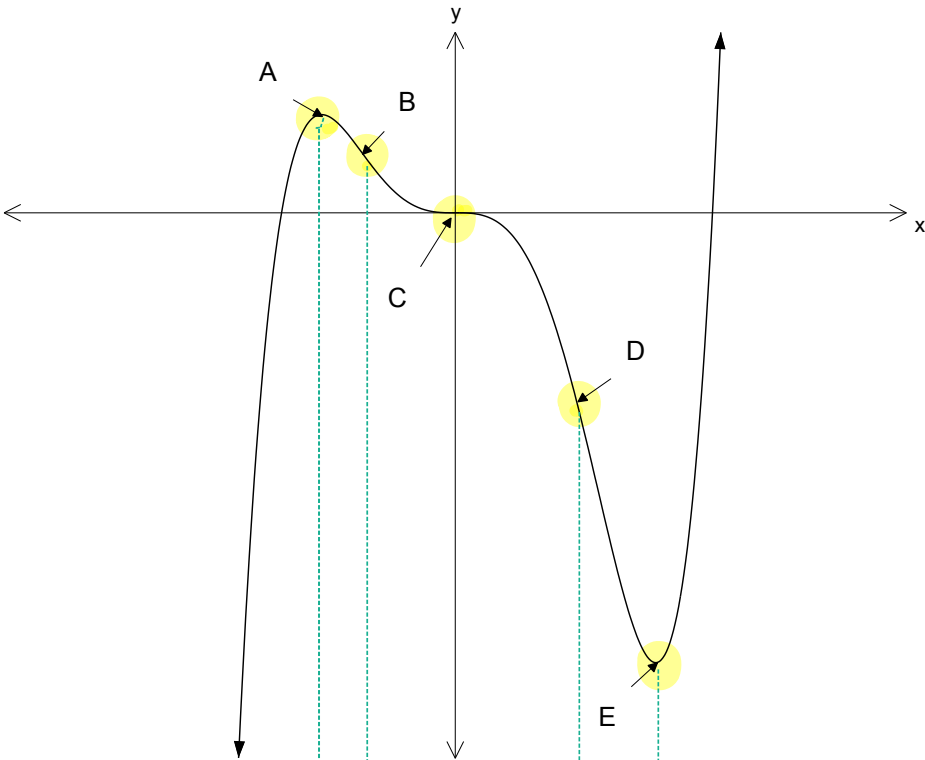
For how many days did she collect mushrooms?

$$\begin{aligned}
 &S_n = 3220 \\
 &\frac{n}{2} [2a_1 + (n-1)d] = 3220 \\
 &\frac{n}{2} [2(34) + (n-1) \cdot 6] = 3220 \quad (1) \\
 &34n + 3n^2 - 3n = 3220 \\
 &3n^2 + 31n - 3220 = 0 \\
 &n = 28 \quad \text{or} \quad n = -\frac{115}{3} \quad (1) \\
 &\qquad \qquad \text{not valid as } n > 0 \\
 &\therefore n = 28 \text{ days} \quad (1)
 \end{aligned}$$

Question 21 (2 marks)

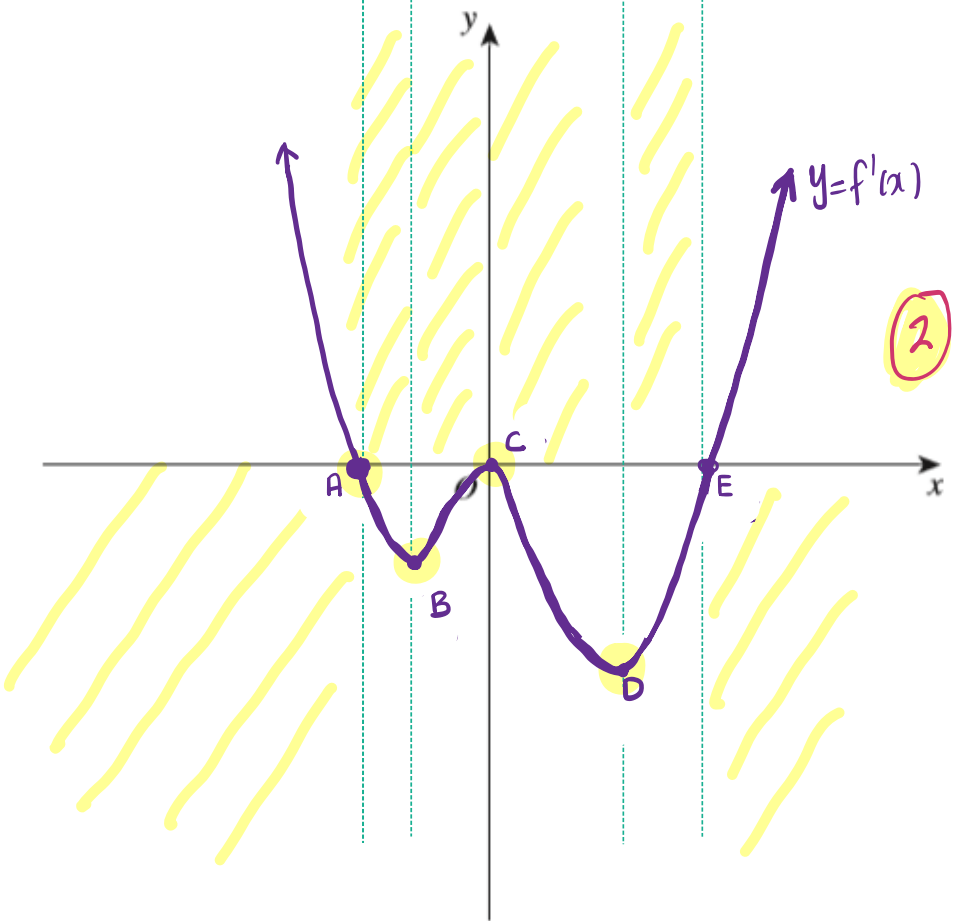
The graph of the curve  $y = f(x)$  is drawn below:

2



Inf pt  $\rightarrow$  tp  $\rightarrow$  x-int  
hor inf  $\rightarrow$  tp & x-int

On the axes below, sketch the graph of  $y = f'(x)$ , Clearly showing the behaviour of the points: A,B,C,D and E.



**Question 22** (7 marks)

The temperature,  $T$  degrees, of a liquid is given according to the model  $T(t) = 20 + Ae^{kt}$ , where  $t$  is time in hours. The initial temperature of the liquid is 35 degrees.

- (a) Show that  $A = 15$ .

1

$$t=0, T(0)=35$$

$$35 = 20 + A \cdot e^{k \times 0}$$

$$35 = 20 + A$$

$$15 = A$$

$$\therefore A = 15$$

- (b) After four hours, the temperature of the liquid is 80 degrees.

3

Show that  $k = \ln(\sqrt{2})$ .

$$t=4, T(4)=80, k=?$$

$$80 = 20 + 15 \cdot e^{k \cdot 4}$$

$$60 = 15 e^{k \cdot 4}$$

$$4 = e^{4k}$$

$$\ln 4 = 4k$$

$$k = \frac{\ln 4}{4}$$

$$k = \frac{1}{4} \ln 2^2$$

$$k = \frac{1}{2} \ln 2$$

$$k = \ln \sqrt{2}$$

Question 22 continues on page 20

## Question 22 (continued)

- (c) Show that  $T(t) = 15 \cdot 2^{\frac{t}{2}} + 20$  and hence sketch its graph.

3

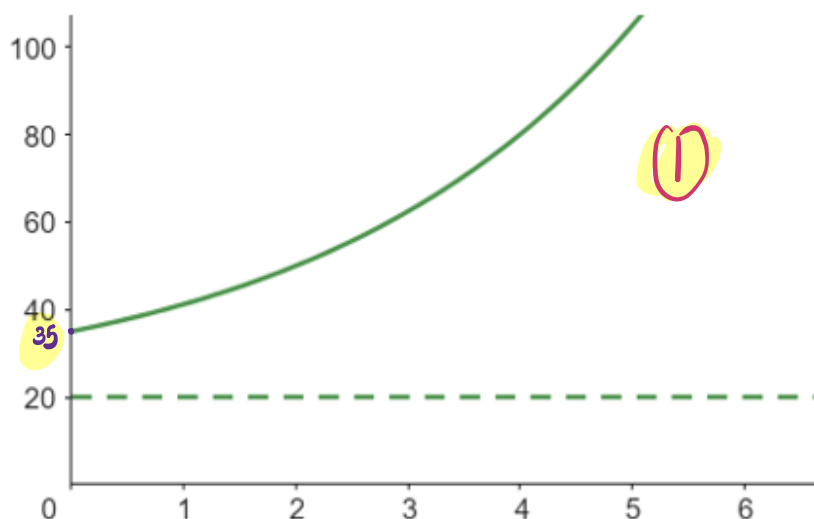
$$T(t) = 20 + 15 e^{\ln \sqrt{2} \cdot t}$$

$$T(t) = 20 + 15 (e^{\ln \sqrt{2}})^t$$

$$= 20 + 15 \cdot (\sqrt{2})^t$$

$$= 20 + 15 \cdot 2^{\frac{t}{2}}$$

$$\therefore T(t) = 15 \cdot 2^{\frac{t}{2}} + 20$$

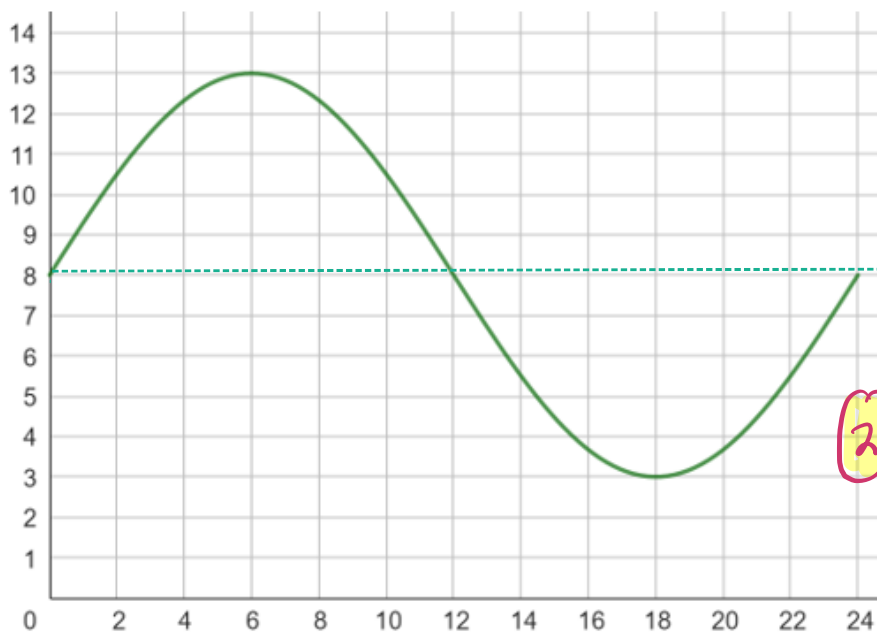


End of Question 22

**Question 23** (9 marks)

The water at a certain beach rises and falls in a periodic pattern. The height of the water at any time can be modelled by the equation  $h(t) = 8 + 5 \sin\left(\frac{\pi t}{12}\right)$ , where  $t$  is time in hours after midnight.

- (a) Sketch the graph of  $y = h(t)$  for  $0 \leq t \leq 24$ .



$$\text{Period} = \frac{2\pi}{\frac{\pi}{12}}$$

$$= 24$$

$$8 \uparrow$$

$$\text{max} = 13$$

$$\text{min} = 3$$

- (b) At what rate does the height of water change at 8:00 am? Leave your answer in exact form.

$$h = 8 + 5 \sin\left(\frac{\pi t}{12}\right)$$

$$h' = 5 \cdot \frac{\pi}{12} \cdot \cos\left(\frac{\pi}{12} t\right)$$

$$h' = \frac{5\pi}{12} \cos\left(\frac{\pi}{12} t\right) \quad (1)$$

$$\text{at } 8:00 \text{ am} \rightarrow t = 8$$

$$\begin{aligned} h'(8) &= \frac{5\pi}{12} \cdot \cos\left(\frac{\pi}{12} \times 8\right) \\ &= \frac{5\pi}{12} \cdot \cos\left(\frac{2\pi}{3}\right) \end{aligned}$$

$$h'(8) = \frac{5\pi}{12} \cdot \left(-\frac{1}{2}\right)$$

$$h'(8) = -\frac{5\pi}{24} \quad (1)$$

Question 23 continues on page 22

## Question 23 (continued)

- (c) A boat with a depth of 5.5 metres, is scheduled to arrive at the beach at 10:00 am one day.

2

Explain if the boat will be able to dock.

$$h(10) = 8 + 5 \sin\left(\frac{5\pi}{6}\right)$$

$$= 10.5$$

$$10.5 > 5.5 \quad \text{So Yes able to dock.}$$

- (d) On each day, between what times will the boat in part (c) be able to dock?

3

$$8 + 5 \sin\left(\frac{\pi}{12}t\right) = 5.5$$

$$5 \sin\left(\frac{\pi}{12}t\right) = -2.5$$

$$\sin\left(\frac{\pi}{12}t\right) = -\frac{1}{2}$$

$$\left(\sin \frac{\pi}{6} = \frac{1}{2} \quad \theta = \frac{\pi}{6}\right)$$

| S                | A                 |
|------------------|-------------------|
| T                | C                 |
| $(\pi + \theta)$ | $(2\pi - \theta)$ |

$$\frac{\pi}{12}t = \left(\pi + \frac{\pi}{6}\right), \left(2\pi - \frac{\pi}{6}\right)$$

$$\frac{\pi}{12}t = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$t = 14, 22$$

$$\therefore t = 2\text{pm or } 10\text{pm}$$

$$\therefore t < 2\text{pm} \quad t > 10\text{pm}$$

before 2pm or after 10pm.

End of Question 23

## Question 24 (3 marks)

3

Find the value of  $k$  such that  $\int_0^k e^x(e^x - 2)dx = \frac{3}{2}$ .

$$\int_0^k (e^{2x} - 2e^x) dx = \frac{3}{2}$$

$$\left[ \frac{1}{2}e^{2x} - 2e^x \right]_0^k = \frac{3}{2}$$

$$\left[ \frac{1}{2}e^{2k} - 2e^k \right] - \left[ \frac{1}{2}e^{2 \cdot 0} - 2e^0 \right] = \frac{3}{2}$$

$$\left[ \frac{1}{2}e^{2k} - 2e^k \right] - \left[ -\frac{3}{2} \right] = \frac{3}{2}$$

$$\frac{1}{2}e^{2k} - 2e^k = 0$$

$$e^{2k} - 4e^k = 0$$

$$e^k(e^k - 4) = 0$$

$$\therefore e^k = 0 \quad (1) \quad \text{or} \quad e^k = 4$$

no sol<sup>n</sup>

$$k = \ln 4$$

$$\text{or } k = \ln 2^2$$

$$\therefore k = 2 \ln 2.$$

**Question 25** (4 marks)

A continuous random variable  $X$  has probability density function given by:

$$f(x) = \begin{cases} \frac{x+2}{k} & 1 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

(a) Show that  $k = 8$ .

1

$$\int_1^3 \frac{x+2}{k} dx = 1$$

$$\frac{1}{k} \int_1^3 (x+2) dx = 1$$

$$\left[ \frac{x^2}{2} + 2x \right]_1^3 = k \quad \textcircled{1}$$

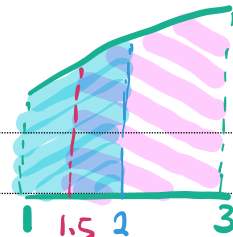
$$\left[ \frac{9}{2} + 6 \right] - \left[ \frac{1}{2} + 2 \right] = k$$

$$8 = k$$

(b) Find  $P(X < 2 \mid X > 1.5)$ .

3

$$P(X < 2 \mid X > 1.5) = \frac{P(X < 2) \cap P(X > 1.5)}{P(X > 1.5)}$$



$$= \frac{P(1.5 < X < 2)}{P(1.5 < X < 3)} = \frac{\left[ \frac{1}{2}x^2 + 2x \right]_{1.5}^2}{\left[ \frac{1}{2}x^2 + 2x \right]_{1.5}^3} \quad \textcircled{1} \quad \textcircled{1}$$

$$= \frac{\int_{1.5}^2 (x+2) dx}{\int_{1.5}^3 (x+2) dx} = \frac{\left( \frac{1}{2} \cdot 2^2 + 2 \cdot 2 \right) - \left( \frac{1}{2} \cdot 1.5^2 + 2 \cdot 1.5 \right)}{\left( \frac{1}{2} \cdot 3^2 + 2 \cdot 3 \right) - \left( \frac{1}{2} \cdot 1.5^2 + 2 \cdot 1.5 \right)}$$

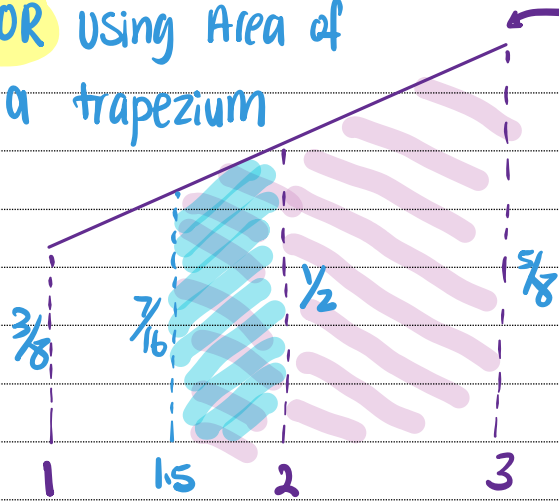
$$= \frac{3}{17} \quad \textcircled{1}$$

End of Section 11 – Booklet II

## Section II extra writing space

If you use this space, clearly indicate which question you are answering.

OR Using Area of  
a trapezium



$$\frac{1}{8}(x+2)$$

$$x=1, \frac{1}{8}(1+2) = \frac{3}{8}$$

$$x=1.5, \frac{1}{8}(1.5+2) = \frac{7}{16}$$

$$x=2, \frac{1}{8}(2+2) = \frac{1}{2}$$

$$x=3, \frac{1}{8}(3+2) = \frac{5}{8}$$

$$\text{using Area} = \frac{h}{2}(a+b)$$

$$\frac{P(1.5 < x < 2)}{P(1.5 < x < 3)} = \frac{\frac{0.5}{2} \left( \frac{1}{2} + \frac{7}{16} \right)}{\frac{1.5}{2} \left( \frac{7}{16} + \frac{5}{8} \right)}$$

$$= \frac{\frac{15}{64}}{\frac{51}{64}} = \frac{5}{17}$$

**2023**

**HSC TRIAL  
EXAMINATION**

Student Name: \_\_\_\_\_

## **Mathematics Advanced**

### **Section II**

### **Answer Booklet - III**

**30 marks**

**Attempt Questions 26–31**

**Allow about 60 minutes for this booklet**

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**Instructions**

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
  - Your responses should include relevant mathematical reasoning and/or calculations.
  - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
  - Write your name above
- 

**Please turn over**

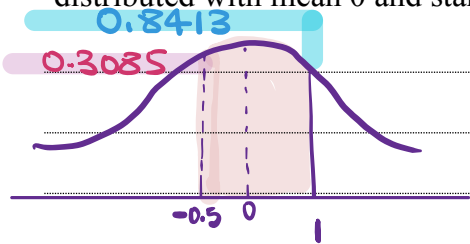
**Question 26** (5 marks)

A random variable is normally distributed with mean 0 and standard deviation 1.

The table gives the probability that this random variable is less than  $z$  for different values of  $z$ .

| $z$        | -2     | -1.5   | -1     | -0.5   | 0      | 0.5    | 1      | 1.5    | 2      |
|------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| $P(Z < z)$ | 0.0228 | 0.0668 | 0.1587 | 0.3085 | 0.5000 | 0.6915 | 0.8413 | 0.9332 | 0.9772 |

- (a) Using the table, find the probability that a value from a random variable that is normally distributed with mean 0 and standard deviation 1, lies between -0.5 and 1. 1



$$0.8413 - 0.3085 = 0.5328$$

①

- (b) The blood pressure readings of patients with a certain condition are normally distributed, with a mean of  $\mu$  and a standard deviation of  $\sigma$ . It is known that 2.28% of these patients have a blood reading of less than 105 and 6.68% of these patients have a blood reading of more than 133. 4

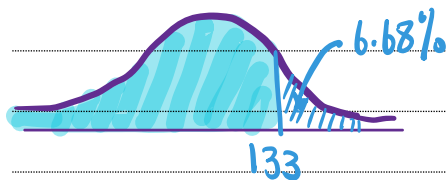
Using the table, find  $\mu$  and  $\sigma$ .

$$2.28\% = 0.0228 \rightarrow P(Z < -2) = 0.0228$$

$$\frac{105 - \mu}{\sigma} = -2 \quad \text{①}$$

6.68% more than 133

$$105 - \mu = -2\sigma \quad \text{--- ①}$$



$$\frac{133 - \mu}{\sigma} = 1.5 \quad \text{①}$$

$$100\% - 6.68\% = 93.32\% = 0.9332$$

which gives  $z = 1.5$

② - ①

$$133 - \mu = 1.5\sigma \quad \text{--- ②}$$

$$28 = 3.5\sigma$$

$$8 = \sigma \quad \text{①}$$

subst  $\sigma = 8$  into ①

$$105 - \mu = -2(8)$$

$$\mu = 121 \quad \text{①}$$

## Question 27 (4 marks)

Find the global maximum and minimum values of  $f(x) = \sin^2 x + 4 \cos x$ , where  $\frac{\pi}{3} \leq x \leq \frac{3\pi}{2}$ .

4

$$f(x) = \sin^2 x + 4 \cos x$$

$$f'(x) = 2 \sin x \cdot \cos x - 4 \sin x \quad (1)$$

$$f'(x) = 0 \rightarrow \text{stat points}$$

$$0 = 2 \sin x \cos x - 4 \sin x$$

$$0 = 2 \sin x (\cos x - 2)$$

$$\downarrow$$

$$2 \sin x = 0$$

$$\sin x = 0$$

$$x = 0, \pi, 2\pi$$

$$\downarrow$$

$$\cos x - 2 = 0$$

$$\cos x = 2$$

no sol<sup>n</sup>

$$(1) \quad D: \frac{\pi}{3} < x < \frac{3\pi}{2}$$

$$\therefore x = \pi$$

$$f\left(\frac{\pi}{3}\right) = \frac{11}{4}$$

$$f(x) \text{ max} = \frac{11}{4} \quad (1)$$

$$f(\pi) = -4$$

$$f(x) \text{ min} = -4 \quad (1)$$

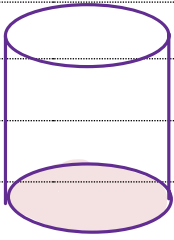
$$f\left(\frac{3\pi}{2}\right) = 1$$

**Question 28** (5 marks)

A container in the shape of a right circular cylinder with no top, has outside surface area  $12\pi$  square metres.

- (a) Show that the height ( $h$ ) is  $\frac{12-r^2}{2r}$ , where  $r$  is the radius.

1



$$SA = \pi r^2 + 2\pi r h$$

$$12\pi = \pi r^2 + 2\pi r h$$

$$12\pi - \pi r^2 = 2\pi r h$$

$$h = \frac{12\pi - \pi r^2}{2\pi r}$$

$$h = \frac{\cancel{\pi}(12-r^2)}{\cancel{2\pi}r} = \frac{12-r^2}{2r} \quad (1)$$

- (b) Find and verify the height ( $h$ ) and radius ( $r$ ) that give the maximum volume.

4

$$V = \pi r^2 h$$

$$V = \pi \cdot r^2 \cdot \frac{12-r^2}{2r}$$

$$(1) \quad V = \frac{\pi}{2} (12r - r^3)$$

$$\frac{dV}{dr} = \frac{\pi}{2} (12 - 3r^2)$$

$$0 = \frac{\pi}{2} (12 - 3r^2) \quad (1)$$

$$3r^2 = 12$$

$$r^2 = 4$$

$$r = \pm 2$$

$$r > 0$$

$$\therefore r = 2$$

$$V'' = \frac{\pi}{2} (-6r)$$

$$V''(2) = \frac{\pi}{2} (-6 \cdot 2) = -6\pi < 0 \quad \curvearrowright \text{max}$$

$$\therefore r = 2 \quad (1)$$

$$h = \frac{12-2^2}{2 \cdot 2}$$

$$h = \frac{12-4}{4}$$

$$h = 2 \quad (1)$$

$\therefore$  Max volume is given by

$$r = 2, \quad h = 2$$

## Question 29 (4 marks)

In the diagram, the region between the graphs of  $y = \frac{4}{x}$  and  $y = \sqrt{2x}$  for  $1 \leq x \leq 4$  is shaded.

4

P.O.I.:

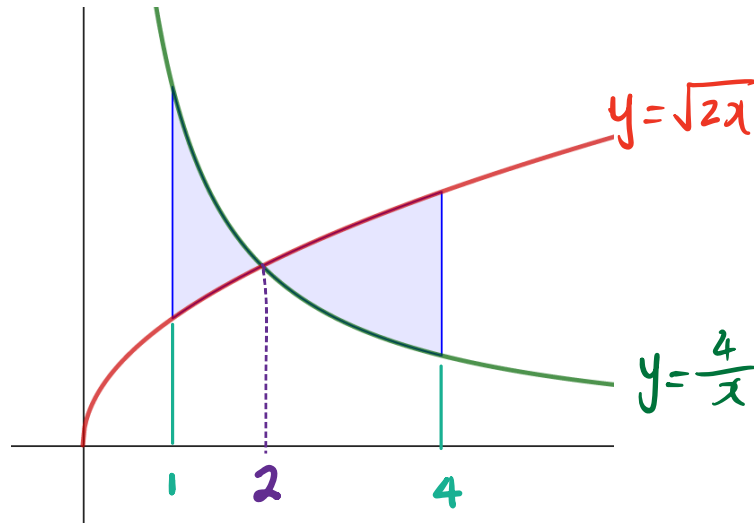
$$\sqrt{2x} = \frac{4}{x}$$

$$2x = \frac{16}{x^2}$$

$$2x^3 = 16$$

$$x^3 = 8$$

$$x = 2 \quad \textcircled{1}$$



Find the exact area of the shaded region.

$$A_1 = \int_1^2 \left( \frac{4}{x} - \sqrt{2} x^{\frac{1}{2}} \right) dx$$

$$= \left[ 4 \ln|x| - \frac{\sqrt{2} x^{\frac{3}{2}}}{\frac{3}{2}} \right]_1^2$$

$$= \left[ 4 \ln 2 - \frac{2\sqrt{2}}{3} (2)^{\frac{3}{2}} \right] - \left[ 4 \ln 1 - \frac{2\sqrt{2}}{3} (1)^{\frac{3}{2}} \right]$$

$$= \left( 4 \ln 2 - \frac{8}{3} \right) + \frac{2\sqrt{2}}{3} \quad \textcircled{1}$$

$$= 4 \ln 2 - \frac{8}{3} + \frac{2\sqrt{2}}{3}$$

$$A_2 = \int_2^4 \left( \sqrt{2} x^{\frac{1}{2}} - \frac{4}{x} \right) dx$$

$$= \left[ \frac{2\sqrt{2}}{3} x^{\frac{3}{2}} - 4 \ln|x| \right]_2^4 \quad \textcircled{1}$$

$$= \left( \frac{2\sqrt{2}}{3} (4)^{\frac{3}{2}} - 4 \ln 4 \right) - \left( \frac{2\sqrt{2}}{3} \cdot 2^{\frac{3}{2}} - 4 \ln 2 \right)$$

$$= \frac{16\sqrt{2}}{3} - 8 \ln 2 - \frac{8}{3} + 4 \ln 2$$

$$A_1 + A_2 = 4 \ln 2 - \frac{8}{3} + \frac{2\sqrt{2}}{3} + \frac{16\sqrt{2}}{3} - 8 \ln 2 - \frac{8}{3} + 4 \ln 2$$

$$= \frac{18\sqrt{2}}{3} - \frac{16}{3} = 6\sqrt{2} - \frac{16}{3}$$

**Question 30** (8 marks)

Eman invests \$9000 in an account earning interest at a rate of 0.6% per month. Each month, immediately after the interest has been paid, Eman contributes \$200.

The amount in the account after the  $n$ th contribution can be determined using the recurrence relation  $A_n = 1.006A_{n-1} + 200$ , where  $A_0 = 9000$ .

- (a) Find the amount in the account after the first contribution.

1

$$\begin{aligned} A_1 &= A_0 \cdot r + 200 \\ A_1 &= 9000(1.006) + 200 \\ A_1 &= 9254 \end{aligned}$$

- (b) Find the total value of the investment after the 12<sup>th</sup> contribution.

3

$$\begin{aligned} A_2 &= A_1(1.006) + 200 \\ &= 9000(1.006)^2 + 200(1.006) + 200 \\ A_3 &= 9000(1.006)^3 + 200(1.006)^2 + 200(1.006) + 200 \\ &\vdots \\ A_{12} &= 9000(1.006)^{12} + 200(1.006)^{11} + \dots + 200 \\ A_{12} &= 9000(1.006)^{12} + 2000 \left[ (1.006)^{11} + \dots + 1.006 + 1 \right] \\ &= 9000(1.006)^{12} + 2000 \left( \frac{1.006^{12} - 1}{1.006 - 1} \right) \\ &= 12150.60 \end{aligned}$$

Question 30 continues on page 32

## Question 30 (continued)

- (c) After the 24<sup>th</sup> contribution the total value of the investment is \$15 536. At that time, Eman stops contributing and starts to withdraw \$450 every following month. The balance in the account will still earn interest at a rate of 0.6% per month.

4

How many complete months will Eman be able to withdraw \$450 from this account ?

$$A_{24} = 15536$$

$$\text{Let: } B_0 = 15536, M = \$450, r = 1.006, B_n = 0, n = ?$$

$$B_1 = B_0 r - M$$

$$B_2 = B_0 r^2 - M r - M$$

$$\vdots$$

$$B_n = B_0 r^n - M r^{n-1} - \dots - M$$

} (1)

$$0 = 15536 (1.006)^n - 450 (1 + 1.006 + \dots + 1.006^{n-1})$$

$$450 \times \frac{(1.006^n - 1)}{1.006 - 1} = 15536 (1.006)^n \quad (1)$$

$$75000 (1.006)^n - 75000 = 15536 (1.006)^n$$

$$59464 (1.006)^n = 75000$$

$$(1.006)^n = \frac{75000}{59464}$$

$$n = \ln\left(\frac{75000}{59464}\right) \div \ln(1.006)$$

$$n \div 38.8 \quad (1)$$

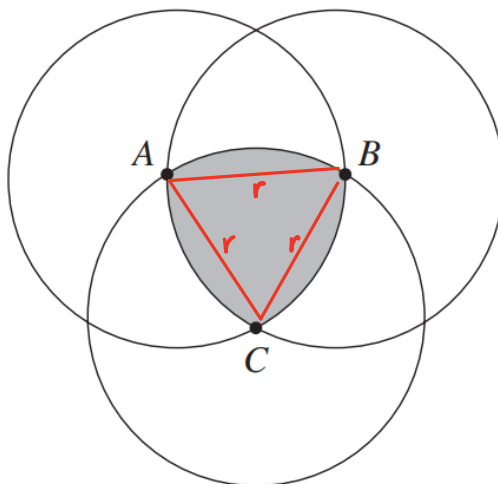
$$n = 38 \text{ months.} \quad (1)$$

End of Question 30

**Question 31** (4 marks)

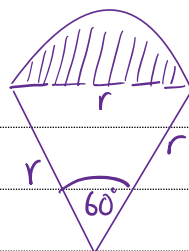
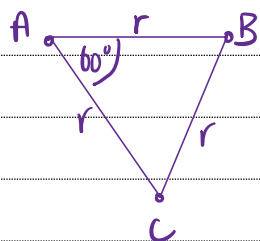
Three circles with radius  $r$  are drawn such that their centres lie on the circumference of the other two circles, as shown in the diagram.

4



Points  $A$ ,  $B$  and  $C$  are the centres of the three circles. Prove that the area of the shaded region is

$$\frac{1}{2}r^2(\pi - \sqrt{3}).$$



$$A = \frac{1}{2}ab \sin C$$

$$= \frac{1}{2} \cdot r \cdot r \sin 60^\circ$$

$$= \frac{1}{2} r^2 \cdot \frac{\sqrt{3}}{2}$$

$$= \frac{r^2 \sqrt{3}}{4}$$

(1)

$$A = \frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta$$

$$= \frac{1}{2}r^2\left(\frac{\pi}{3} - \sin\frac{\pi}{3}\right)$$

$$= \frac{1}{2}r^2\left(\frac{\pi}{3} - \frac{\sqrt{3}}{2}\right)$$

$$= \frac{\pi r^2}{6} - \frac{\sqrt{3}r^2}{4}$$

(1)

$$\text{Total Area} = \frac{r^2 \sqrt{3}}{4} + 3 \times \left( \frac{\pi r^2}{6} - \frac{\sqrt{3} r^2}{4} \right)$$

$$= \frac{r^2 \sqrt{3}}{4} + \frac{\pi r^2}{2} - \frac{3\sqrt{3} r^2}{4}$$

$$= \frac{\pi r^2}{2} - \frac{\sqrt{3} r^2}{4}$$

$$= \frac{r^2}{2} (\pi - \sqrt{3})$$

(1)

**End of EXAM**

## Section II extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

## Section II extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Mathematics Advanced Yr 12

Section I Answer Sheet

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response circle completely.

|    |                         |                         |                         |                         |
|----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| 2  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| 3  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| 4  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| 5  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| 6  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| 7  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| 8  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| 9  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| 10 | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |

Mathematics Advanced  
Mathematics Extension 1  
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3} Ah$$

$$V = \frac{4}{3} \pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For  $ax^3 + bx^2 + cx + d = 0$ :

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{a(1 - r^n)}{1 - r}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

## Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2} ab \sin C$$

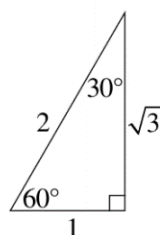
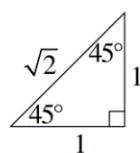
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2} r^2 \theta$$



## Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

## Compound angles

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A+B) - \sin(A-B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

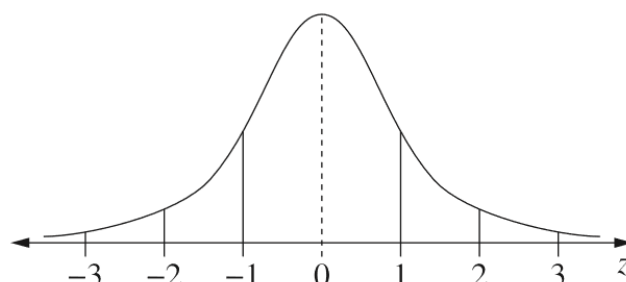
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

## Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score  
less than  $Q_1 - 1.5 \times IQR$   
or  
more than  $Q_3 + 1.5 \times IQR$

## Normal distribution



- approximately 68% of scores have z-scores between  $-1$  and  $1$
- approximately 95% of scores have z-scores between  $-2$  and  $2$
- approximately 99.7% of scores have z-scores between  $-3$  and  $3$

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

## Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

## Continuous random variables

$$P(X \leq x) = \int_a^x f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

## Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= {}^nC_x p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

## Differential Calculus

### Function

### Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = nf'(x)[f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u), u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

## Integral Calculus

$$\int f'(x)[f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where  $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx \approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\}$$

where  $a = x_0$  and  $b = x_n$

## Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \dots + \binom{n}{r}x^{n-r}a^r + \dots + a^n$$

## Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}||\underline{v}|\cos\theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda\underline{b}$$

## Complex Numbers

$$z = a + ib = r(\cos\theta + i\sin\theta) \\ = re^{i\theta}$$

$$\left[r(\cos\theta + i\sin\theta)\right]^n = r^n(\cos n\theta + i\sin n\theta) \\ = r^n e^{in\theta}$$

## Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

